

The magic of flight

THE IDEA OF FLIGHT, and the extraordinary freedom it bestows, is as old as the human imagination itself. The desire to take to the air was there from the start. Anyone who climbed a hill had an idea of the vista that might unfold, the wonders they might see—if only they were able to emulate the birds. The Montgolfier brothers launched the gentle art of ballooning in magnificent style in 1783 and Sir George Cayley sent the world's first—albeit reluctant—glider pilot off for a one-off swoop across a Yorkshire valley in 1853. However, for all the technological advances of the Industrial Revolution, by the end of the 19th century the dream of sustained, controllable flight remained unfulfilled.

The appearance of the gasoline engine suggested that the goal was within grasp, but the unsolved problem of control—or lack of it—was taking its toll. The courageous German pioneer Otto Lilienthal was killed when his glider stalled and crashed—an unfortunate demonstration of the limitations of a pilot attempting to steer an aircraft by shifting his body weight. “Sacrifices,” the dying man said to his brother “have to be made.” It was the improbable genius of brothers Orville and Wilbur Wright, cycle makers from Dayton, Ohio, that finally unlocked the door. Understanding that it was essential to master control, they combined the idea of twisting an aircraft's wings in order to roll it, with an invention of their own: an interconnected rudder that prevented the aircraft from skidding into turns.

Over the course of three years of research in their spare time, the Wrights not only built one of the world's first wind tunnels and tested hundreds of airfoil sections but also established reliable lift and drag data for the design of their first powered machine. Finding that the science of propeller design was virtually nonexistent, they came up with their own theories, and made propellers that were as efficient as those used on light aircraft today. In this way the Wright brothers came to make the world's first sustained and controlled heavier-than-air flight on December 17, 1903—inventing the airplane and giving the US an early lead in aviation.

Word was slow to spread at first, but sufficient detail of the Wright brothers' achievements and—critically—their methods spread around the globe and affected progress elsewhere. Further inspired by Wilbur Wright's masterful 1908 flight demonstrations at Le Mans, the French surged ahead. By 1912 the airspeed record had been raised to more than 100 mph (161km/h) by the intrepid Jules Védrines, flying a rotary-engined Deperdussin monoplane that was far more advanced than anything the US could then produce.

Once the genie was out of the bottle, progress was astonishing—as the pages that follow will reveal. By the end of World War I, Fokker had developed in Germany the combination of welded-steel tube fuselage and cantilever wings that we still use today. In the 1920s and 30s the airliner

